

MA 301
Test 1
Spring - 2015

Problem	points	out of
1		8
2		6
3		6
4		3
5		3
6		2
7		2
Score		30

Name: SOLUTIONS

HR: _____

Signature: _____

READ BEFORE YOU START

- * Show all work.
- * Change your calculator mode to RADIANS.
- * Sloppy solutions may be penalized.
- * Use calculators ONLY for computations.
- * You must write up every problem in a clear and organized way.
- * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Given $F = [2, 2y + z, y + 1]$

(a) Show that F is conservative in two different ways

1. $\text{curl } \vec{F} = [0, 0, 0]$ ✓

2. $[2, 2y + z, y + 1] = [f_x, f_y, f_z]$

$$f_x = 2$$

$$f = 2x + g(y, z)$$

$$2y + z = f_y = g_y$$

$$\Rightarrow g_y = 2y + z$$

$$\Rightarrow g = y^2 + zy + h(z)$$

$$f = 2x + y^2 + zy + h(z)$$

$$y + 1 = f_z = y + h_z$$

$$\Rightarrow h_z = 1$$

$$\Rightarrow h = z + C$$

$$f(x, y, z) = 2x + y^2 + zy + z \quad \leftarrow \text{there exists a potential function for } \vec{F} \quad \checkmark$$

(b) Assuming F is a force, compute the work required to move a mass from $(-1, 4, 2)$ to $(0, 0, -6)$

$$W = \int_C \vec{F} \cdot d\vec{r}$$

C is any path
between these
points

$$= f(0, 0, -6) - f(-1, 4, 2)$$

$$= -6 - (-2 + 16 + 8 + 2)$$

$$= -30$$

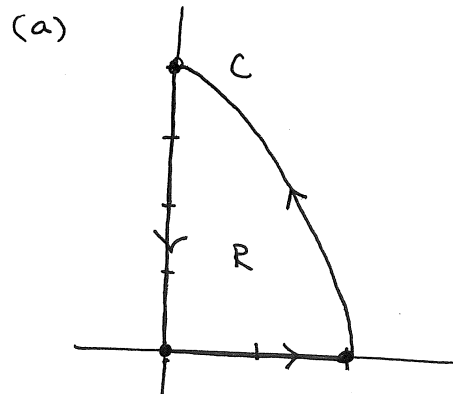
$$\vec{F}(x,y) = [xy - x^{15}, 2x + \ln y]$$

Problem 2: Suppose $\vec{F}(x,y) = [xy - x^{15}, 2x + \ln y]$ and suppose C consists of the following 3 curves: (i) the line segment from $(0,0)$ to $(2,0)$, (ii) the curve $y = 4 - x^2$ from $(2,0)$ to $(0,4)$ and (iii) the line segment from $(0,4)$ to $(0,0)$.

(a) Sketch C

(b) Compute $\int_C \vec{F} \cdot d\vec{r}$

$$N_x - M_y = 2 - x$$



$$\begin{aligned}
 (b) \quad \int_C \vec{F} \cdot d\vec{r} &= \iint_R (2 - x) \, dA \\
 &= \int_0^2 \int_0^{4-x^2} (2-x) \, dy \, dx \\
 &= \int_0^2 (2-x)y \Big|_0^{4-x^2} \, dx \\
 &= \int_0^2 (2-x)(4-x^2) \, dx \\
 &= \int_0^2 (8 - 4x - 2x^2 + x^3) \, dx \\
 &= \left[8x - 2x^2 - \frac{2}{3}x^3 + \frac{1}{4}x^4 \right]_0^2 \\
 &= 8(2) - 2(2)^2 - \frac{2}{3}(2)^3 + \frac{1}{4}(2)^4 - 0 \\
 &= 16 - 8 - \frac{16}{3} + 4 \\
 &= 12 - \frac{16}{3} \\
 &= \frac{36}{3} - \frac{16}{3} = \frac{20}{3} \approx 6.667
 \end{aligned}$$

Problem 3: Let $\mathbf{F} = [z^2, x - y, 7]$ and C be the intersection of the surfaces $x = y + z$ and $y = z^2$ with $0 \leq y \leq 9$.

Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$

$$z = t \quad \mathbf{r}(t) = [t^2 + t, t^2, t] \quad t \in [0, 3]$$

$$\mathbf{r}'(t) = [2t + 1, 2t, 1]$$

$$\mathbf{F}(\mathbf{r}(t)) = [t^2, t^2 + t - t^2, 7]$$

$$= [t^2, t, 7]$$

$$\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = 2t^3 + t^2 + 2t^2 + 7$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^3 2t^3 + 3t^2 + 7 \, dt$$

$$= \left. \frac{1}{2}t^4 + t^3 + 7t \right|_0^3$$

$$= \frac{81}{2} + 27 + 21 - 0$$

$$= \frac{81}{2} + 48$$

$$= \frac{81}{2} + \frac{96}{2} = \frac{177}{2} = 88.5$$

Problem 4: Answer the following true/false questions

- T (a) True or False: If C is an ellipse and F is irrotational then $\int_C F \cdot dr = 0$
- T (b) True or False: If $F = \nabla f$ then F is conservative.
- T (c) True or False: Line integrals over all constant vector fields are path-independent.

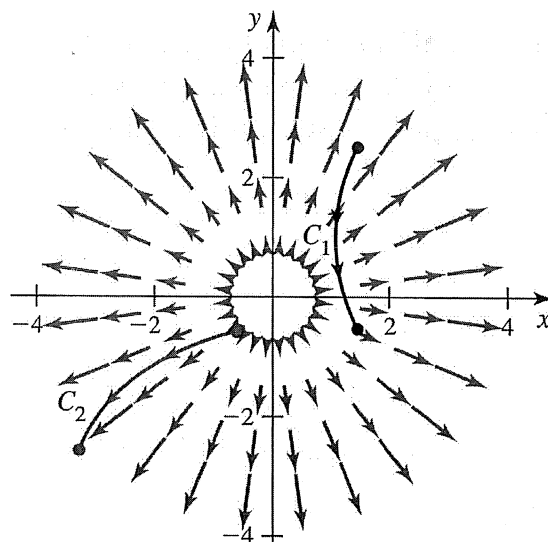
Problem 5: Let F be the vector field and C_1 and C_2 be the curves shown in the provided figure.

- (a) Would the work for a mass traveling along C_1 be positive, negative, or zero? Explain.

Negative

- (b) Would the work for a mass traveling along C_2 be positive, negative, or zero? Explain.

Positive



- (c) Which is larger: $\|\text{curl}(F(2,2))\|$ or $\|\text{div}(F(2,2))\|$? Explain.
- $\downarrow$ 0 $\downarrow$ positive

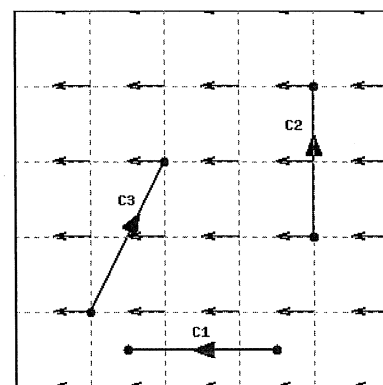
Problem 6: Given the figure, answer the following:

- (a) List the line integrals $\int_{C_1} F \cdot dr$, $\int_{C_2} F \cdot dr$, $\int_{C_3} F \cdot dr$ in increasing order value.

$$\int_{C_3} \vec{F} \cdot d\vec{r}, \quad \int_{C_2} \vec{F} \cdot d\vec{r}, \quad \int_{C_1} \vec{F} \cdot d\vec{r}$$

- (b) Circle the integrals that are path independent

$\int_{C_1} F \cdot dr$ $\int_{C_2} F \cdot dr$ $\int_{C_3} F \cdot dr$



Problem 7: Let $F = \nabla(\ln x + \underbrace{(x-y)^2}_f)$ and C be parameterized by $r(t) = [e^t, t^2 - t]$ where $0 \leq t \leq 2$.

Compute $\int_C F \cdot dr = f(B) - f(A)$

$$\begin{aligned} &= \ln(e^2) + (e^2 - 2)^2 - (\ln(1) + (1-0)^2) \\ &= 2 + (e^2 - 2)^2 - 1 \\ &= 1 + (e^2 - 2)^2 \approx 30.042 \end{aligned}$$

$$\begin{aligned} A &= \vec{r}(0) = [1, 0] \\ B &= \vec{r}(2) = [e^2, 2] \end{aligned}$$